

C# Final Report Assignment

Instructions for Students:

Prepare final report for the following questions given in assignments and lab works for visual programming with c#.

Include questions , code snippets and screenshots of output of each questions in your report .

For Final Project you can include

- Question
- Project description
- Screenshots of the interface and output
- Code snippets of major components
- Database schema
- List of features implemented
- Learnings form your final project and further enhancement ideas .

Chapter One: Basic Programming Concepts

1. Write a C# program to swap two numbers using a third variable.
2. Write a C# program to swap two numbers without using a third variable.
3. Write a C# program to input a number from the user and check if it is a prime number.
4. Write a C# program that checks whether a number is a palindrome.
5. Write a C# program to reverse a number entered from the console.
6. Write a C# program to check whether a number is an Armstrong number.
7. Write a C# program to convert temperature from Fahrenheit to Celsius.
8. Write a C# program to print the first N numbers of the Fibonacci series using recursion.
9. Write a C# program to find the second largest number in an array.
10. Write a C# program to count the number of vowels (a, e, i, o, u) in a given string.
11. Write a C# program to find the missing number in an array of integers from 1 to N.

Chapter Two: Strings and Lambda Expressions

12. Write a C# program to reverse a string entered by the user.
13. Write a C# program to remove duplicate characters from a string.
14. Write a lambda expression in C# to sort the following list of strings by length:
`List<string> words = new List<string> { "apple", "dog", "banana" };`
15. Write a lambda expression in C# to check if there are any negative numbers in the list:
`List<int> data = new List<int> { 1, 2, -3, 4 };`
16. Write a lambda expression to filter even numbers from a list.

17. Write a lambda expression to check if a given string starts with the letter 'A'.

Chapter Three: LINQ Queries

18. Write a LINQ query to get the first 3 elements of the list:

[10, 20, 30, 40, 50]

19. Write a LINQ query to find all strings longer than 5 characters in a list.

20. Write a LINQ query to join two lists based on matching Student ID and display:

- Student Name
- Course
- Grade

Student List:

- Id: 1, Name: Alice
- Id: 2, Name: Bob

Grade List:

- StudentId: 1, Course: Math, Grade: A
- StudentId: 2, Course: Science, Grade: B

Chapter Four: Database and ADO.NET

Use the following table structure:

Student Table

Column Name	DataType
StudentId	int
Name	varchar
Age	int

21. Create a C# class (model) for the Student table.

22. Write a SQL query to create the Student table in the database.

23. Write a stored procedure in SQL to insert student records.

24. Write an ADO.NET method to insert a student into the table using the stored procedure:

```
public void InsertStudent(int studentId, string name, int age)
{
    // Your ADO.NET code to insert data into the Students table
}
```

Final Project: CRUD Application with Login

Title: _____

Project Requirements:

- Implement a login system
- Perform Create, Read, Update, Delete (CRUD) operations on Employee records
- Use Windows Forms or ASP.NET with ADO.NET or Entity Framework